CSE 242 Assignment 3

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1. SVM

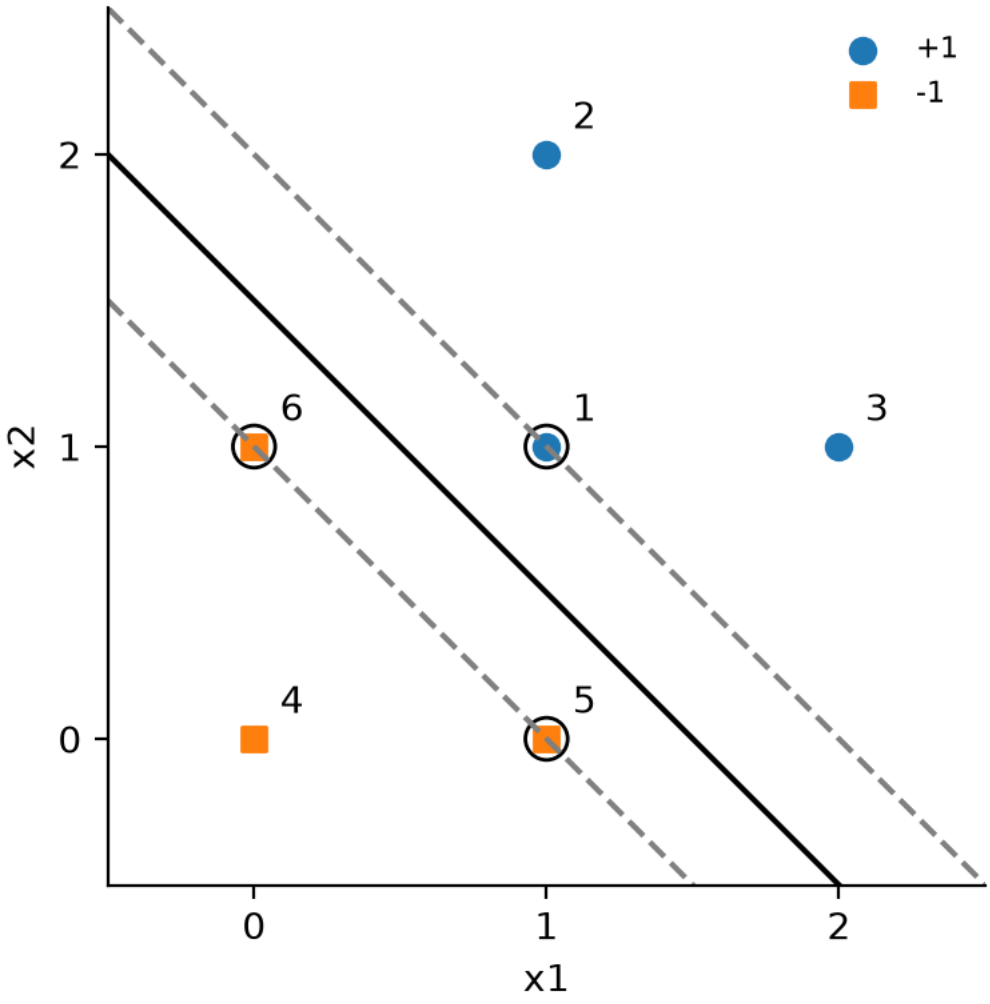
SVs: 1, 5, 6, on the closest supporting lines.

$$w_1 + w_2 + b = 1, \quad w_1 + b = w_2 + b = -1$$

$$w = (2, 2),\ b = -3 \quad \Rightarrow \quad 2x_1 + 2x_2 - 3 = 0$$

$$\text{margin} = 1/\|w\| = 1/\sqrt{8} \approx 0.3536$$

closest hull points: (1,1), (0.5,0.5). The line bisects their gap.

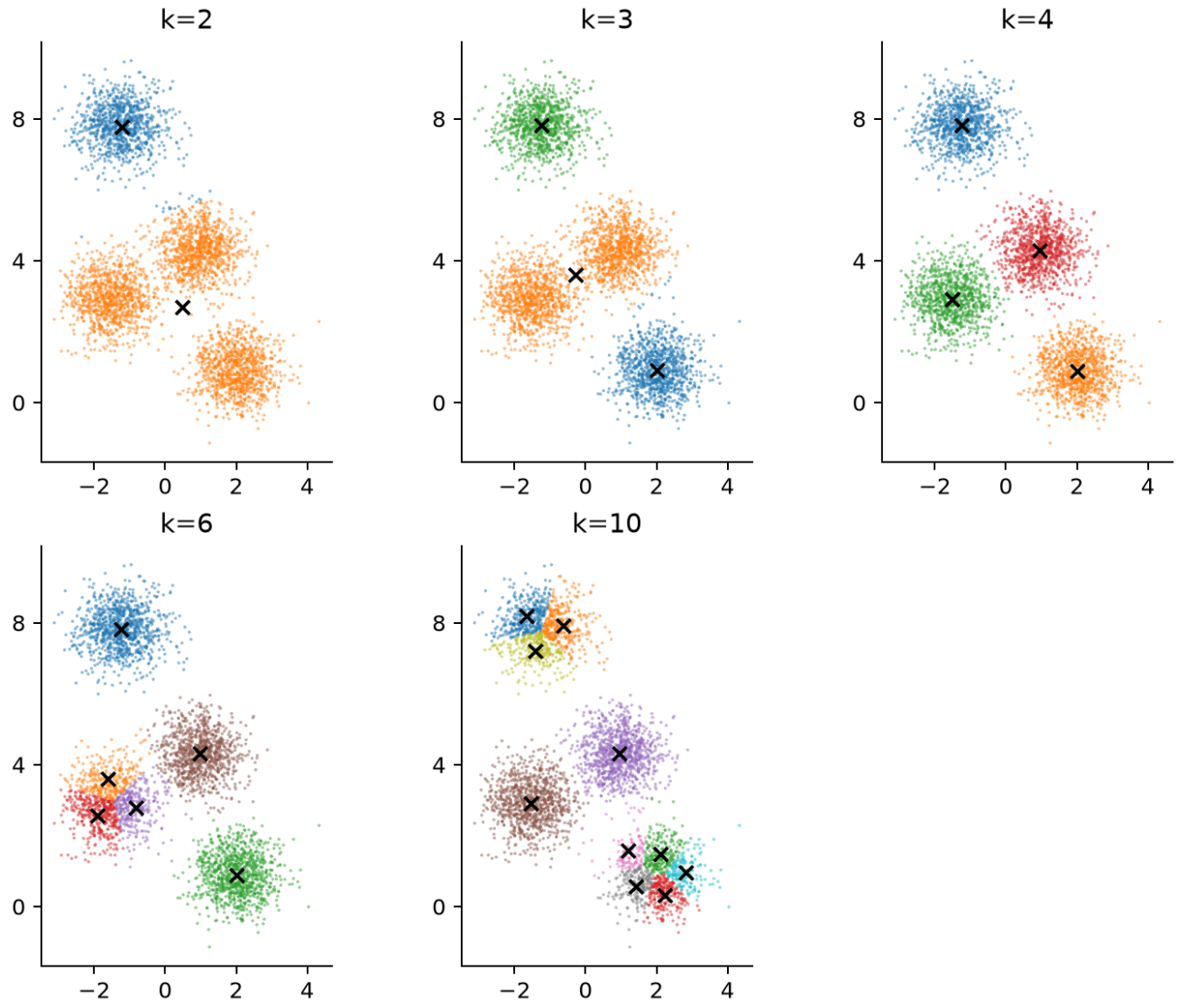


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2. K-means

ties: lowest cluster index; empty clusters: keep old center. x = centroid.



k=2,3 merge groups; k=4 matches the blobs; k=6,10 split them.

3. Noisy labels

seed 0; 20 trials; 5 other neighbors, synchronous votes. ties: row index.

Baseline	K-NN repair	Change
82.80%	86.44%	+3.64 pp

helped here: local votes remove flips, but can spread errors near overlap.